

1.Second Largest element in Array

```
public class Main {  
    public static void main(String[] args) {  
  
        int arr[] = {10, 25, 8, 40, 30};  
  
        int big = 0;  
        int second = 0;  
  
        for (int i = 0; i < arr.length; i++) {  
  
            if (arr[i] > big) {  
                second = big; // old big becomes second  
                big = arr[i]; // new big  
            }  
            else if (arr[i] > second) {  
                second = arr[i];  
            }  
        }  
  
        System.out.println("Second Highest = " + second);  
    }  
}
```

2. First Largest element in Array.

```
public class Main {  
    public static void main(String[] args) {  
  
        int arr[] = {10, 25, 8, 40, 30};  
  
        int big = arr[0]; // assume first element is largest  
  
        for (int i = 1; i < arr.length; i++) {  
  
            if (arr[i] > big) {  
                big = arr[i]; // update largest  
            }  
        }  
  
        System.out.println("Largest = " + big);  
    }  
}
```

3. Validation {}

```
boolean isBalanced(String s) {
    Stack<Character> st = new Stack<>();

    for (char c : s.toCharArray()) {

        if (c == '(' || c == '{' || c == '[') {
            st.push(c);
        }

        else if (c == ')' || c == '}' || c == ']') {

            if (st.isEmpty()) return false;

            char top = st.peek();

            if ((c == ')' && top == '(') ||
                (c == '}' && top == '{') ||
                (c == ']' && top == '[')) {
                st.pop();
            } else {
                return false;
            }
        }
    }

    return st.isEmpty();
}
```

3. Star Pattern

```
*
***
*****
*****
*****
```

```
public class Main {
    public static void main(String[] args) {

        int n = 5;

        for (int i = 1; i <= n; i++) {
```

```

        for (int space = 1; space <= n - i; space++) {
            System.out.print(" ");
        }

        for (int star = 1; star <= 2 * i - 1; star++) {
            System.out.print("*");
        }

        System.out.println();
    }
}

```

4. Binary Search

```

public class Main {
    public static void main(String[] args) {

        int arr[] = {10, 20, 30, 40, 50};
        int key = 30;

        int start = 0;
        int end = arr.length - 1;

        while (start <= end) {

            int mid = (start + end) / 2;

            if (arr[mid] == key) {
                System.out.println("Found at index: " + mid);
                break;
            }
            else if (key > arr[mid]) {
                start = mid + 1;
            }
            else {
                end = mid - 1;
            }
        }
    }
}

```

—5. Frequence—

```

public class Main {

```

```
public static void main(String[] args) {  
  
    String str = "banana";  
    char ch = 'a';  
  
    int count = 0;  
  
    for (int i = 0; i < str.length(); i++) {  
  
        if (str.charAt(i) == ch) {  
            count++;  
        }  
    }  
  
    System.out.println("Frequency = " + count);  
}  
}
```